

In this paper, all exercises in Hoffman/Kunze will be solved in complete form.

$$A = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & & & \\ \vdots & & & \\ A_{n1} & \dots & \dots & A_{nn} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} B_{11} & B_{12} & \dots & B_{1n} \\ B_{21} & & & \\ \vdots & & & \\ B_{n1} & \dots & \dots & B_{nn} \end{pmatrix}$$

Denote $e_j \in F^{n \times 1}$, then

$$Ae_j = \begin{pmatrix} A_{1j} \\ A_{2j} \\ \vdots \\ A_{nj} \end{pmatrix} \quad \text{and} \quad e_i^T A = (A_{i1} \ A_{i2} \ \dots \ A_{in})$$

$$AB = \left(\sum_{k=1}^n A_{ik} \cdot B_{kj} \right)_{i,j} = \left(f(A^T e_i, A e_j) \right)_{i,j},$$

where $f: F^n \times F^n \rightarrow F: \left(\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \right) \mapsto \sum_{k=1}^n x_k y_k$, which is the bilinear form on F^n .

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}_{n \times 1} \cdot \begin{bmatrix} b_1 & b_2 & \dots & b_n \end{bmatrix}_{1 \times n} = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \dots & a_1 b_n \\ a_2 b_1 & & & \\ \vdots & & a_{ij} b_j & \\ \vdots & & & \\ a_n b_1 & \dots & \dots & a_n b_n \end{bmatrix}$$

Sec 3.1. Linear Transformation.

#3. Let $V = F[x]$, and $D: V \rightarrow V$ be a formal differentiation.

Find kernel and range of D .

Proof If the characteristic of F is 0, then

$$Df = 0 \Leftrightarrow f \text{ is constant, thus } \ker D = F.$$

And, the basis for V is $\{x^k \mid k=0,1,2,\dots\}$, so $Dx^k = k \cdot x^{k-1}$ ($k \geq 1$) implies

$$\operatorname{Im} D = V.$$

If the characteristic of F is not zero, a prime p , then

$$Dx^n = 0 \Leftrightarrow n \in (p)$$

$$\ker D = \operatorname{span}_F \{x^{pk} \mid k=0,1,2,\dots\}$$

And

$$\operatorname{Im} D = \operatorname{span}_F \{x^{n-1} \mid n \in \mathbb{N} \text{ and } n \notin (p)\}$$

#6. Describe explicitly a linear operator $T: F^2 \rightarrow F^2$ s.t.

$$T(e_1) = \begin{pmatrix} a \\ b \end{pmatrix} \text{ and } T(e_2) = \begin{pmatrix} c \\ d \end{pmatrix}$$

$$\text{Proof. } [T]_{\{e_1, e_2\}} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \text{ or } T(x, y) = (ax + cy, bx + dy)$$

#8. Find a linear $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ s.t. $\operatorname{Im} T = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\}$

$$\text{Proof Take } [T]_{\{e_1, e_2, e_3\}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ -1 & 2 & 2 \end{pmatrix}.$$

#9. Verify for fixed $B \in M_{n \times n}(F)$,

$$T: M_{n \times n}(F) \rightarrow M_{n \times n}(F); A \mapsto AB - BA$$

is linear map.

Proof. Given $A_1, A_2 \in M_{n \times n}(F)$,

$$T(A_1 + A_2) = (A_1 + A_2)B - B(A_1 + A_2)$$

$$= A_1B - BA_1 + A_2B - BA_2$$

$$= T(A_1) + T(A_2). \quad \text{And clearly } T(cA) = c(AB - BA) = cT(A) \text{ for any } c \in F.$$

#10. Find a map on $\mathbb{C}$ to $\mathbb{C}$ is $\mathbb{R}$ -linear but not $\mathbb{C}$ -linear.

Proof. $T: \mathbb{C} \rightarrow \mathbb{C}; a+bi \mapsto a$.

Given $a+bi, c+di \in \mathbb{C}$,

$$T(a+bi + c+di) = a+c = T(a+bi) + T(c+di),$$

so, it preserves the addition. And, for any real number $r \in \mathbb{R}$,

$$T(r \cdot a + r \cdot bi) = ra = r \cdot T(a+bi)$$

But for $i \in \mathbb{C}$,

$$T(i \cdot a + i \cdot bi) = T(-b + ai) = -b \neq i \cdot T(a+bi) = ia.$$



#11. Let $A \in M_{m \times n}(F)$.

Prove that $T: F^n \rightarrow F^m; X \mapsto AX$

is zero map if and only if A is zero matrix.

Proof. T is zero map $\Leftrightarrow$ for any $X \in F^n$, $T(X) = 0$
 $\Leftrightarrow$ for each $1 \leq i \leq n$, $T(e_i) = 0$
 $\Leftrightarrow$ " $Ae_i = 0$
 $\Leftrightarrow$ all rows of A are zero
 $\Leftrightarrow A = 0$.

#13. Let $T: V \rightarrow V$ be a linear operator. Then,

$\text{Im } T \cap \text{Ker } T = \{0\}$ if and only if $T(T(\alpha)) = 0 \Rightarrow T(\alpha) = 0$.

Proof. Since $T(\alpha) \in \text{Im } T$ for all $\alpha \in V$, if $T(T(\alpha)) = 0 \Rightarrow T(\alpha) \in \text{Ker } T \Rightarrow T(\alpha) = 0$.
Conversely, for any $\beta \in \text{Im } T \cap \text{Ker } T$, $\beta = T(\alpha)$ for some α , and, if $\beta = 0$ by the assumption.

Sec 3.2.

#3, $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3: (x_1, x_2, x_3) \mapsto (3x_1, x_1 - x_2, 2x_1 + x_2 + x_3)$

is invertible?

Proof $T(e_1) = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$ $T(e_2) = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$ $T(e_3) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

So $[T]_{\mathcal{B}} = \begin{pmatrix} 3 & 0 & 0 \\ 1 & -1 & 0 \\ 2 & 1 & 1 \end{pmatrix}$ is invertible, $\det T = -3$.

#6. Let $T: F^3 \rightarrow F^2$ and $U: F^2 \rightarrow F^3$ be linear maps

prove that $U \circ T: F^3 \rightarrow F^3$ is not invertible

Proof. $U \circ T$ invertible $\Leftrightarrow U \circ T$ is onto, by the dimension theorem.

But, $U \circ T[F^3] = U[F^2] \neq F^3$, since $\dim F^2 = 2 < 3 = \dim F^3$.

#7. Find $T, U: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ s.t. $TU = 0$ but $UT \neq 0$.

Proof. Let $T = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $U = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$,

$TU = 0$, but $UT = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0$.

#8. Find $T^2 = 0$ but $T \neq 0$. $T = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

#9. Let V be a fin. dim space, and $T, U: V \rightarrow V$ be linear operators s.t. $TU = I$.
Then, T is invertible with $T^{-1} = U$.

Find an Example it is false when V is not fin. dim.

Proof. Since $(T \circ U)[V] = V$, $U[V] = V$, so U is onto.

By the Dimension Theorem, $\ker U = \{0\}$, so U is invertible.

Now, $T = U^{-1}$.

#11. Let V be a fin. dim vector space and $T: V \rightarrow V$ be a linear operator.

If $\text{rank } T^2 = \text{rank } T$, then $\text{Ker } T \cap \text{Im } T = \{0\}$. (So $\text{Ker } T \oplus \text{Im } T = V$).

Proof. Since for any $\alpha \in V$, $T(T(\alpha)) \in \text{Im } T^2$, so $\text{Im } T \subseteq \text{Im } T^2$.

And, the dimensions are same by assertion, $\text{Im } T = \text{Im } T^2$.

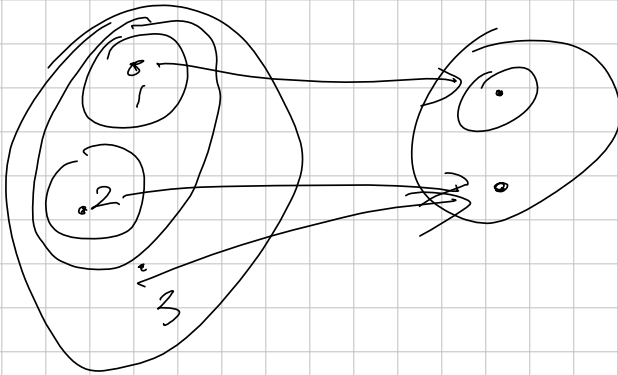
Moreover, if $T(\alpha) = 0$, then $T(T(\alpha)) = 0$, so $\text{Ker } T \subseteq \text{Ker } T^2$.

and, similarly, $\text{Ker } T = \text{Ker } T^2$.

Now, let $\beta \in \text{Ker } T \cap \text{Im } T$. Then, $\beta = T(\alpha)$ for some $\alpha \in V$ and $T(\beta) = T^2(\alpha) = 0$.

$T^2(\alpha) = 0$ implies $\alpha \in \text{Ker } T^2$, so $\alpha \in \text{Ker } T$. That is, $\beta = T(\alpha) = 0$.

So proved.



$$\{\{1\}, \{2\}, \{1, 2\}, X, \emptyset\}$$

$$\{\{1\}, Y, \emptyset\}$$

$$[0, 2\pi] \rightarrow \mathbb{C} : t \mapsto e^{it}$$

$$[0, \pi) \rightarrow X$$

$$\begin{array}{l} \pi : \mathbb{R}^2 \rightarrow \mathbb{R} : (x, y) \mapsto x \\ \hline \{ \frac{1}{n}, n \} \quad \{ (\frac{1}{n}, n) \mid n \in \mathbb{N} \} \end{array}$$

Sec 3.3. Let V and W be vector spaces over F , and $U: V \rightarrow W$ be an Isomorphism.

Prove that $L(V, V) \rightarrow L(W, W): T \mapsto UTU^{-1}$

is an isomorphism.

Proof. Given $T_1, T_2 \in L(V, V)$ and $c \in F$,

$$U(cT_1 + T_2)U^{-1} = cUT_1U^{-1} + UT_2U^{-1}$$

so this map is linear. And,

$$UT_1U^{-1} = UT_2U^{-1} \Rightarrow U^{-1}UT_1U^{-1}U = U^{-1}UT_2U^{-1}U \Rightarrow T_1 = T_2$$

thus injective. And, given $S \in L(W, W)$,

$$U^{-1}SU \in L(V, V) \text{ satisfies } UU^{-1}SUU^{-1} = S.$$

Sec. 3.5 linear functional.

#3.5.3. For any $A, B \in M_{n \times n}(F)$, $\text{trace}(AB) = \text{trace}(BA)$.

Moreover, if $A \sim B$, then $\text{trace } A = \text{trace } B$.

Proof. Since $\text{trace } AB = \sum_{j=1}^n \sum_{k=1}^n A_{jk} B_{kj}$ and $\text{trace } BA = \sum_{j=1}^n \sum_{k=1}^n B_{jk} A_{kj}$

so just rename the indices, we obtain the first assertion.

Now, suppose that $A = U^{-1}BU$ for some $U \in GL(F)$. Now,

$$\text{trace } A = \text{trace } U^{-1}BU = \text{trace } BUU^{-1} = \text{trace } B.$$

#5.2.3,

Given a Matrix $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ over the commutative Ring with identity K ,

a Matrix $\text{adj } A = \begin{bmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{bmatrix}$ is called the classical adjoint of A .

Using the properties of $\det$, n -linear, alternating, $\det I = 1$, deduce the following:

a). $(\text{adj } A) \cdot A = A \cdot (\text{adj } A) = (\det A) \cdot I$.

b). $\det(\text{adj } A) = \det A$

c). $\text{adj}(A^t) = (\text{adj } A)^t$.

Proof.

Sec 6.2.

6.2.1. Note f_2 is the characteristic polynomial of the given matrix A_2 .

$$f_1(t) = t(t-1), \quad f_2(t) = (2-t)/(1-t) + 3 = t^2 - 3t + 5, \quad f_3(t) = (1-t)^2 - 1 = t^2 - 2t$$

In $\mathbb{R}$, f_2 and f_3 do not split over the given field, actually there are no solution in $\mathbb{R}$ of f_2 and f_3 . So A_2 and A_3 have no eigenvalues.

But, $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $E_0 = \text{Ker } A = \text{span}_{\mathbb{R}} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

And $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, so $E_1 = \text{span}_{\mathbb{R}} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$, $\frac{2 \pm \sqrt{4-8}}{2}$

In $\mathbb{C}^2$, A_1 has same eigenvectors. But:

$$t = \frac{3 \pm \sqrt{9-20}}{2} = \frac{3 \pm i\sqrt{11}}{2} \text{ are roots of } f_2, \text{ and}$$

$$t = \frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i \text{ are roots of } f_3. \text{ so,}$$

$$A_2 - \underbrace{\left(\frac{3 \pm i\sqrt{11}}{2} \right)}_{\lambda_2} I = \begin{bmatrix} \frac{1}{2} \pm i \frac{\sqrt{11}}{2} & 3 \\ -1 & -\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \end{bmatrix}$$

$$\text{Then, } (A_2 - \lambda_2)I \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \left(\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right)x + 3y \\ -x + \left(-\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right)y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} \left(\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right)x + 3y = 0 \\ \left(\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right)x + 3y = 0. \end{cases}$$

(multiplying $\left(\frac{1}{2} - i \frac{\sqrt{11}}{2} \right)$ to the bottom)

so, the eigenvectors are:

$$\begin{pmatrix} -1 \\ \frac{1}{3} \left(\frac{1}{2} \pm i \frac{\sqrt{11}}{2} \right) \end{pmatrix}, \text{ corresponding to } \frac{1}{2} \pm i \frac{\sqrt{11}}{2}$$

#6.2.6. $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{bmatrix}$ over $\mathbb{R}$.

$\det(A - tI) = t^4$, and A has the only eigenvalue as 0. So,

A is diagonalizable iff $\dim \ker A = 4$.

But, the only possibility for $\dim \ker A = 4$ is $a=b=c=0$, because if at least one of a or b or c is not zero, then $\text{rank } A \geq 1$.

#6.2.7. If $T: F^n \rightarrow F^n$ has n distinct eigenvalues, then T is diagonalizable.

Proof Let $\alpha_1, \dots, \alpha_n \in F^n$ be the eigenvectors corresponding to distinct eigenvalues. Then, $\{\alpha_1, \dots, \alpha_n\}$ are linearly independent, because: Suppose that

$$c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n = 0 \text{ where } c_i \in F.$$

Then, for any F -polynomial $f(t) \in F[t]$,

$$0 = f(T)(c_1 \alpha_1 + \dots + c_n \alpha_n) = f(c_1) \cdot \alpha_1 + \dots + f(c_n) \alpha_n. \quad (*)$$

$$\text{Take } f_i(t) = \prod_{j \neq i} (t - c_j) / \prod_{j \neq i} (c_i - c_j)$$

Then, $f_i(c_i) = 1$, so, we can conclude $c_1 = c_2 = \dots = c_n = 0$,

by substituting $f_1, \dots, f_n$ to $(*)$.

#6.2.8. Let $A, B \in M_n(F)$. If $I - AB$ is invertible, then $I - BA$ is invertible with

$$(I - BA)^{-1} = I + B \cdot (I - AB)^{-1} \cdot A$$

Proof. Let $C = (I - AB)^{-1}$. Then,

$$C - ABC = C - CAB = I, \text{ so } ABC = CAB \text{ (That is, } C \text{ and } AB \text{ commute.)}$$

$$\begin{aligned} \text{Now } (I - BA) \cdot (I + BCA) &= I + BCA - BA - BABC A \\ &= I + BCA - BA - B(C - I)A \\ &= I + BCA - BA - BCA + BA \\ &= I. \end{aligned}$$

So, proved.

#6.2.9. Using the fact above, deduce that

AB and BA have precisely the same eigenvalue in F .

Proof. If $t = 0$, then

$$t \text{ is an eigenvalue for } AB \Leftrightarrow \det AB = 0 \Leftrightarrow \det BA = 0 \Leftrightarrow t \text{ is an eigenvalue for } BA.$$

$$\begin{aligned} \text{If } t \neq 0, \text{ then } \det(AB - tI) = 0 &\Leftrightarrow t^n \cdot \det((AB) \cdot t^{-1} - I) = 0 \\ &\Leftrightarrow \det((BA) \cdot t^{-1} - I) = 0 \\ &\Leftrightarrow \det(BA - tI) = 0. \end{aligned}$$

So, proved.

#6.2.10. A 2×2 symmetric matrix is diagonalizable.

Proof. Let $A = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$ be given. Then

$$\det(A - tI) = (t-a)(t-b) - c^2 = t^2 - (a+b)t + ab - c^2$$

If $c=0$, then it is clearly diagonal. And, if $c \neq 0$, then $c^2 > 0$, so the characteristic polynomial has two distinct roots r_1 and r_2 ,

$$\dim \ker(A - r_1 I) \geq 1 \quad \text{and} \quad \dim \ker(A - r_2 I) \geq 1,$$

so $\dim \ker A - r_1 I = \dim \ker A - r_2 I = 1$, so diagonalizable.

#6.2.11. If $N \in M_{\text{real}}(\mathbb{C})$ satisfies $N^2 = 0$, then either $N=0$ or $N \sim \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

Proof. If $N=0$, then we are done. Suppose that $N \neq 0$, but $N^2=0$.

(Directly, the minimal polynomial of N is t^2 , so it can be triangular.

But in this, we prove more specifically.)

Since $N^2=0$ implies $N \cdot Nx = 0$ for all $x \in \mathbb{C}^2$, so

$$\text{Im } N \subseteq \ker N.$$

Thus $\ker N \neq \{0\}$, take $d_1 \in \ker N$ be non-zero.

And, since $N \neq 0$, $\ker N \neq \mathbb{C}^2$. So take $d_2 \in \mathbb{C}^2 \setminus \ker N$ s.t. $Nd_2 \in \ker N$.

Now, $Nd_1 = 0$ and $Nd_2 = C \cdot d_1$ for some non-zero scalar $C \in \mathbb{F}$, so

$$[L_N]_{\{d_1, d_2\}} = \begin{bmatrix} 0 & C \\ 0 & 0 \end{bmatrix} \quad \text{or} \quad [L_N]_{\{d_2, d_1\}} = \begin{bmatrix} 0 & 0 \\ C & 0 \end{bmatrix}.$$

(To conclude the conversation completely, take $d_2' = C^{-1}d_2$).

#6.2.12. Every Complex 2×2 Matrix is similar to $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ or $\begin{bmatrix} a & 1 \\ 0 & a \end{bmatrix}$.

Proof. The characteristic polynomial of the given Matrix must be degree 2 complex polynomial.

So, there are only two possibilities: the polynomial has only one root of multiplicity 2

or distinct two roots.

In the second case, it is diagonalizable. In the first case, let $C \in \mathbb{F}$ be the root.

Then, $(N - CI)^2 = 0$, because $\det(N - tI) = (t-C)^2$, so $\ker N - CI \neq \emptyset$.

so $N - CI \sim \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, or $N \sim \begin{bmatrix} a & 1 \\ 0 & a \end{bmatrix}$.

#6.2.13.

Let $V = \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$, and define

$$T: V \rightarrow V; f \mapsto \int_0^x f(t) dt.$$

Then T has no eigenvalues.

Proof. Since the integral has linearity, T is a linear map.

Suppose that T has an eigenvalue, $c \in \mathbb{R}$. That is,

$$T(f) = c \cdot f(x) = \int_0^x f(t) dt \quad \text{for some non-zero } f \in V.$$

If $c=0$, then $\int_0^x f(t) dt = 0$ for all x , so f is zero, contradiction.

If $c \neq 0$, then $c \cdot f(x) = \int_0^x f = 0$. Moreover,

$$c \cdot F'(x) = F(x) \Rightarrow F'(x) = c^{-1} e^{cx} \Rightarrow f(x) = e^{cx}$$

But $e^{c0} = e^0 = 1 \neq 0$, so contradiction.

#6.2.14.

Let $A \in M_{n \times n}(F)$ be a matrix with ch. polynomial

$$(x - c_1)^{d_1} \dots (x - c_k)^{d_k}$$

Let $V = \text{span}_F \{B \in M_{n \times n} \mid AB = BA\}$, then $\dim V = d_1^2 + \dots + d_k^2$.

Proof.

#6.3.2.

Let $a, b, c \in F$. Then,

$$A = \begin{bmatrix} 0 & 0 & -c \\ 1 & 0 & -b \\ 0 & 1 & -a \end{bmatrix} \text{ has characteristic polynomial as } x^3 + ax^2 + bx + c, \text{ and it is minimal polynomial for } A.$$

$$\text{Proof. } \det(xI - A) = \begin{vmatrix} x & 0 & c \\ -1 & x & b \\ 0 & -1 & x+a \end{vmatrix} = x \cdot (x(x+a) + b) + c = x^3 + ax^2 + bx + c.$$

$$\text{And, } A^2 = \begin{bmatrix} 0 & -c & ac \\ 0 & -b & -c+ab \\ 1 & -a & -b+a^2 \end{bmatrix} \text{ and } A^3 = \begin{bmatrix} -c & ac & bc-a^2c \\ -b & -c+ab & ac+b^2-a^2b \\ -a & -b+a^2 & -c+2ab-a^3 \end{bmatrix}$$

So, by the just calculation, $A^3 + aA^2 + bA + cI = 0$.

Now, $\{I, A, A^2\}$ is linearly independent, so there is no polynomial p of degree 2 s.t.

$$p(A) = 0$$

So, the polynomial $x^3 + ax^2 + bx + c$ is minimal polynomial for A .

$$\text{\#6.3.3, put } A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 \\ -2 & -2 & 2 & 1 \\ 1 & 1 & -1 & 0 \end{bmatrix}. \text{ Then, } \det(xI - A) =$$

#6.3.5. Let V be an n -dimensional Vector space and $T: V \rightarrow V$ be a linear operator. If $T^k = 0$ for some $k \in \mathbb{N}$, then $T^n = 0$.

Proof. If $k \leq n$, then clearly, $T^n = T^{n-k} \circ T^k = 0$.

Suppose that $k > n$. By the Cayley-Hamilton Theorem, the minimal polynomial $p(x)$ of T satisfies $\deg p(x) \leq n$, since $p(x)$ divides the characteristic polynomial of T .

So, the minimal polynomial of T also divides T^k , so $T^r = 0$ for some $r \leq n$.

Now, $T^n = T^{n-r} \circ T^r = 0$.

#6.3.6. Find $\begin{smallmatrix} a \\ 3 \end{smallmatrix}$ by 3 Matrix for which the minimal polynomial is x^2 .

Proof. A has minimal polynomial as x^2 if and only if:

$A^2 = 0$ and $\{A, I\}$ is linearly independent.

So, $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

#6.3.7. Let $D: P_n(\mathbb{R}) \rightarrow P_n(\mathbb{R})$ be the differentiation operator.

Evaluate the minimal polynomial for D .

Proof. For any polynomial $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \in P_n(\mathbb{R})$,

$$D^n(\sum a_i x^i) = n! \cdot a_n$$

And, so $D^{n+1} = 0$, the zero map. Now, suppose that

$$C_n D^n + C_{n-1} D^{n-1} + \dots + C_1 D + C_0 I = 0. \text{ where } C_0, \dots, C_n \in \mathbb{R}.$$

Subtracting Polynomials:

$$f_0(x) = 1, f_1(x) = x, f_2(x) = \frac{1}{2!} \cdot x^2, f_3(x) = \frac{1}{3!} \cdot x^3, \dots, f_n(x) = \frac{1}{n!} \cdot x^n$$

Then, the all coefficients are zero, so $\{1, D, \dots, D^n\}$ is linearly independent.

In other say, the polynomial of degree less than $n+1$ cannot be the minimal polynomial.

+Comment for 6.3.7. $\mathbb{R}$ 이 아니라 임의의 체 F 에서 가능, 하지만 $\text{char}(F) = 0$ 이어야 함
 $\mathbb{Z}$ 이 특이한 예외일 것 같으나, $\mathbb{Z}/2\mathbb{Z}$ 이 아.

#6.3.8.

Since $P^2 = I$, the minimal polynomial is $x^2 - x = x(x-1)$.

#6.3.9. If the characteristic polynomial of $A \in M_{n \times n}(F)$ is

$$f = (x - c_1)^{d_1} \cdots (x - c_k)^{d_k},$$

then $\text{trace } A = c_1 d_1 + \cdots + c_k d_k$.

Proof. Since $f = x^n - (c_1 d_1 + \cdots + c_k d_k) \cdot x^{n-1} + \cdots$,

therefore $\text{trace } A = c_1 d_1 + \cdots + c_k d_k$.

(More precisely, characteristic polynomial splits $\Rightarrow$ triangularizable)

And, since the trace is invariant to similar matrix, so we can argue the above statement. Similarly, the determinant is $c_1^{d_1} \cdots c_k^{d_k}$.)

Sec 6.4.

#6.4.1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix} \in M_{2 \times 2}(\mathbb{R})$.

a). The L_A -invariant subspaces are only $\mathbb{R}^2$ and $\{0\}$.

Proof. Let $\{0\} < W < \mathbb{R}^2$ be a L_A -invariant subspace.

Then, $W = \text{span}_{\mathbb{R}}(a, b)$ for some non-zero vector $(a, b) \in \mathbb{R}^2$. So, by the assumption,

$$A \cdot \begin{pmatrix} a \\ b \end{pmatrix} \in W, \text{ or } A \begin{pmatrix} a \\ b \end{pmatrix} = c \cdot \begin{pmatrix} a \\ b \end{pmatrix} \text{ for some } c \in \mathbb{R}.$$

But, $\det(xI - A) = (x-1)(x-2) + 2 = x^2 - 3x + 4 = (x - \frac{3}{2})^2 + \frac{7}{4}$ has no solutions in $\mathbb{R}$, so contradiction.

But, If A is a matrix in $\mathbb{C}$, then it has an eigenvalue α , so $\text{span}_{\mathbb{C}}\{\alpha\}$ is the desired.

#6.4.2. Let W be an invariant space for T .

Then, the minimal polynomial of $T|_W$ divides the minimal polynomial of T .

Proof. Let $P(x)$ be the minimal polynomial of T . Then, for any $\alpha \in W$,

$$P(T)(\alpha) = P(T|_W)(\alpha) = 0.$$

So, $P(T|_W) = 0$, thus we obtain the result.

#6.4.3. Let $c \in F$ be an eigenvalue of T .

For all $\alpha \in \ker(T - cI)$, $T(\alpha) = c \cdot \alpha$, so

$$T|_{\ker(T - cI)} = c \cdot I : \ker(T - cI) \rightarrow \ker(T - cI); \alpha \mapsto c\alpha.$$

So, the restriction is c -multiple of Identity map if $c \neq 0$,
or zero map if $c = 0$.

#6.4.4.

Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 2 & -2 & 2 \\ 2 & -3 & 2 \end{bmatrix}$ over $\mathbb{R}$. Find triangularization for A .

$$\det(xI - A) = \begin{vmatrix} x & -1 & 0 \\ -2 & x+2 & -2 \\ -2 & 3 & x-2 \end{vmatrix} = x((x+2)(x-2) + 6) + (-2)(x-2) - 4$$
$$= x(x^2 + 2) - 2x$$
$$= x^3$$

Put $\alpha_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. Then, $A\alpha_1 = 0$. Now, put $\beta = \begin{pmatrix} 1 \\ 8 \\ 1 \end{pmatrix} \notin \langle \alpha_1 \rangle$. Then, $A\beta = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, $A^2\beta = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \in \langle \alpha_1 \rangle$. So put $\alpha_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

Let $\alpha_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \notin \langle \alpha_1 \rangle \oplus \langle \alpha_2 \rangle$. Then, $A\alpha_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \in \langle \alpha_2 \rangle$. Now, for $P = (\alpha_1 \ \alpha_2 \ \alpha_3) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$, $P^{-1}AP = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

(Actually, it is the Jordan-Cannical form)

#6.4.5. If $A^2 = A$, then A is diagonalizable.

Proof. If $A = 0$, then trivial. Assume $A \neq 0$.

Since $A^2 - A = 0$, the minimal polynomial of A is $x^2 - x$.

So,

$$A \sim \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ where } r = \text{rank } A.$$

#6.4.6. If $T: V \rightarrow V$ is diagonalizable,

then for T -invariant subspace W , $T|_W$ is diagonalizable.

Proof. Since the minimal polynomial of T splits and separable,

and the minimal polynomial of $T|_W$ divides the minimal polynomial of T , so it is also splits and separable.

#6.4.8.

Let $T: V \rightarrow V$ be a linear operator.

If every subspace of V is invariant under T , then T is a scalar multiple of Identity.

Proof. Suppose that there is a vector $\alpha \in V$ s.t. $T(\alpha) \notin \langle \alpha \rangle$.

Then, $\langle \alpha \rangle$ is Not T -invariant, so Contradiction.

So, for any $\alpha \in V$, $T(\alpha) = c \cdot \alpha$ for some $c \in F$.

Let $\alpha_1, \alpha_2 \in V$ be independent. Then, $T(\alpha_1) = c_1 \alpha_1$ and $T(\alpha_2) = c_2 \alpha_2$ for some $c_1, c_2 \in F$.

$$c_1 \alpha_1 + c_2 \alpha_2 = T(\alpha_1) + T(\alpha_2) = T(\alpha_1 + \alpha_2) = c_3 \cdot (\alpha_1 + \alpha_2)$$

By the independence, $c_1 = c_3 = c_2$. Now, for fixed $c \in F$, $T(\alpha) = c\alpha$ for all $\alpha \in F$.

#6.4.9. Let $V = C^0([0,1])$, the space of real-valued continuous map.

Define $T: V \rightarrow V: f \mapsto \int_0^x f(t) dt$.

1). Determine $P[x]$ on $[0,1]$ is T -invariant or Not.

2). Determine $D'[0,1]$ is //

3). Determine $H = \{f \in V \mid f(\frac{1}{2}) = 0\}$.

Proof. Since the integral of polynomial is polynomial, so T -invariant.

And, 2) is also trivial. But 3) cannot T -invariant.

$$f = (x - \frac{1}{2})^2.$$

Comment: $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

#6.4.10.

Let $A \in M_{3 \times 3}(\mathbb{R})$. If A is not similar over $\mathbb{R}$ to a triangular, then A is similar over $\mathbb{C}$ to a diagonal.

Proof. Let $f(t) \in \mathbb{R}[t]$ be the characteristic polynomial of A .

Since $\lim_{t \rightarrow \infty} f(t) = \infty$ and $\lim_{t \rightarrow -\infty} f(t) = -\infty$, $f(t)$ has at least one zero, by the IVT.

But, since A is not similar to a triangular, $f(t)$ does not split over $\mathbb{R}$, it cannot have multiple roots in $\mathbb{C}$, so separable in $\mathbb{C}$. $\square$

#6.4.11. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$. Then, the char. polynomial is

$f(t) = (t-1)(t-2)$, separable, so it is similar to diagonal.

But A is not already diagonal.

#6.4.12. Let F be an algebraically closed field, and V be a fin. dim. v.s. over F . Let $T: V \rightarrow V$ be a linear operator on V , and $f(t) \in F[t]$ be a polynomial.

$c \in F$ is an eigenvalue of $f(T)$ if and only if $c = f(\lambda)$ for some eigenvalue λ of T .

Proof. If $c = f(\lambda)$ for some eigenvalue $\lambda \in F$ corresponding a non-zero vector α , then

$$f(T)(\alpha) = f(\lambda)\alpha = c\alpha, \text{ so } c \text{ is eigenvalue of } f(T).$$

To show the converse, denote $f(t) = (t-c_1) \cdots (t-c_n)$, since F is algebraically closed. Now, $c \in F$ is an eigenvalue of $f(T)$, then for some non-zero vector $\alpha \in V$,

$$c\alpha = f(T)(\alpha) = (T-c_1 I) \circ \cdots \circ (T-c_n I)(\alpha).$$

Conti.

Sec 6.6. Direct-sum.

#6.6.1. Let V be a fin. dim v.s., and W_1 be any subspace of V .

Prove that there is a subspace W_2 of V such that $V = W_1 \oplus W_2$.

Proof. Let $\{\alpha_1, \dots, \alpha_k\}$ be the basis for W_1 . And, extend this to

$\{\alpha_1, \dots, \alpha_k, \alpha_{k+1}, \dots, \alpha_n\}$ being the basis for V .

Now, $W_2 = \text{span}\{\alpha_{k+1}, \dots, \alpha_n\}$ is we desired.

#6.6.2. Let V be a fin. dim v.s and $W_1, \dots, W_k \subseteq V$ be subspaces s.t.

$$V = W_1 + \dots + W_k \quad \text{and} \quad \dim V = \dim W_1 + \dots + \dim W_k.$$

Then, $V = W_1 \oplus \dots \oplus W_k$.

Proof. Let β_i be a basis for W_i , for each $1 \leq i \leq k$. Then,

$\bigcup_{i=1}^k \beta_i$ is the spanning set of V ,

Moreover, by the assumption, $\bigcup_{i=1}^k \beta_i$ is linearly independent.

So, $\{W_1, \dots, W_k\}$ are independent, obtain the result.

#6.6.3. Find the projection on $\mathbb{R}^2$ onto $\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix} \rangle$ along $\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix} \rangle$.

$$E: \mathbb{R}^2 \rightarrow \mathbb{R}^2; \alpha \mapsto \frac{1}{3} \cdot \begin{bmatrix} 2 & -1 \\ -2 & 1 \end{bmatrix} \cdot \alpha.$$

Comment: In General,

let $\{\alpha_1, \dots, \alpha_k\}$ be a basis for $\text{Im } E$ and $\{\alpha_{k+1}, \dots, \alpha_n\}$ be a basis for $\text{Ker } E$.

Then, the invertible Matrix $A = \begin{bmatrix} \alpha_1 & \alpha_2 & \dots & \alpha_n \end{bmatrix}$ satisfies

$$EA = \begin{bmatrix} E\alpha_1 & \dots & E\alpha_n \end{bmatrix} = \begin{bmatrix} \alpha_1 & \dots & \alpha_k & 0 & \dots & 0 \end{bmatrix}$$

Therefore, $E = \begin{bmatrix} \alpha_1 & \dots & \alpha_k & 0 & \dots & 0 \end{bmatrix} \cdot A^{-1} = A \cdot \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} \cdot A^{-1}$ is we desired.

$$\alpha = \alpha_1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \alpha_2 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \begin{pmatrix} 1 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a + b \\ -a + b \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 6 & -3 \\ -6 & 3 \end{pmatrix}$$

$$A^2 = 3A$$

$$\left(\frac{1}{3}A\right)^2 = \frac{1}{3}A^2 = \frac{1}{3}3A$$

$$\left(\frac{1}{3}A\right)^2 = \frac{1}{3}A^2 = \frac{1}{3}3A$$

#6.6.4.

If E_1 and E_2 are Projection such that $\text{Im } E_1$ and $\text{Im } E_2$ are independent, then $E_1 + E_2$ is Projection. (This statement is false!)

Proof. $(E_1 + E_2)^2 = E_1^2 + E_2^2 + E_1 E_2 + E_2 E_1 = E_1 + E_2 + E_1 E_2 + E_2 E_1.$

Therefore $E_1 + E_2$ is projection iff $E_1 E_2 + E_2 E_1 = 0.$

#6.6.5. If E is projection and f is a polynomial, then $f(E) = aI + bE.$

Proof. Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$ Since E is idempotent,

$$f(E) = (a_n + \dots + a_1)E + a_0 I.$$

#6.6.6. False. Counterexample:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \text{ has the characteristic polynomial as } f(t) = (t-1)^2.$$

But, it is Not diagonalizable, because

$$\ker(A - I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}, \dim 1.$$

#6.6.7. Let E be a projection. Then, $I - E$ is a projection on $\ker E$ along $\text{Im } E.$

Proof. First, $(I - E)^2 = I - E - E + E^2 = I - E,$ so $I - E$ is a projection.

Given $\alpha \in V,$ $\alpha \in \text{Im } E \Leftrightarrow \alpha = E(\alpha) \Leftrightarrow (I - E)(\alpha) = 0 \Leftrightarrow \alpha \in \ker(I - E).$
 $\alpha \in \ker E \Leftrightarrow E(\alpha) = 0 \Leftrightarrow (I - E)(\alpha) = \alpha \Leftrightarrow \alpha \in \text{Im}(I - E).$

#6.6.8. Let $E_1, \dots, E_k$ be linear operators on V s.t.

$$E_1 + \dots + E_k = I.$$

a). If $E_i \circ E_j = 0$ for $i \neq j$, then $E_i^2 = E_i$ for each i .

b). In the case $k=2$, prove if $E_1 + E_2 = I$ and $E_1^2 = E_1$, $E_2^2 = E_2$, then $E_1 \circ E_2 = 0$.
Proof.

#6.6.9. Let V be a vector space over $\mathbb{R}$, and E is a projection on V .

Then, $I+E$ is invertible.

Proof. $(I+E) \circ \underbrace{\left(I - \frac{1}{2}E\right)}_{(I+E)^{-1}} = I - \frac{1}{2}E + E - \frac{1}{2}E^2 = I + \frac{1}{2}E - \frac{1}{2}E = I.$

#6.6.10. Let F be a field of Characteristic zero, V be a fin. dim space over F .

Suppose $E_1, \dots, E_k$ are projectors of V s.t. $E_1 + \dots + E_k = I$.

Prove that $E_i \circ E_j = 0$ for $i \neq j$.

Proof. Since every projection on fin. dim space is similar to

$$E_i \sim \begin{bmatrix} I_{r_i} & 0 \\ 0 & 0 \end{bmatrix} \text{ where } r_i = \text{rank } E_i,$$

and trace is invariant under the similar and preserves sum,

$$\dim V = \text{trace } I = \sum \text{trace } E_i = \sum r_i = \sum \dim \text{Im } E_i \dots (!)$$

$$\text{So, } V \stackrel{(*)}{=} \text{Im } E_1 + \dots + \text{Im } E_k = \text{Im } E_1 \oplus \dots \oplus \text{Im } E_k,$$

(*) is obtained by $\alpha = I(\alpha) = (E_1 + \dots + E_k)(\alpha) = E_1(\alpha) + \dots + E_k(\alpha)$ and the direct sum is obtained by (!). Now, given $x \in V$, $x = E_1(x) + \dots + E_k(x)$. So,

$$i \neq j \Rightarrow E_j(x) \in \ker E_i \Rightarrow E_i \circ E_j(x) = 0.$$

#6.7.1.

Let E be a proj. of V , and T be a lin. op. on V .

Prove that:

$\text{Im } E$ is invariant under T if and only if $ETE = TE$

And

$\ker E$ and $\text{Im } E$ are invariant under T if and only if $ET = TE$.

Proof. Suppose that $\text{Im } E$ is invariant under T , that is, $\alpha \in \text{Im } E \Rightarrow T(\alpha) \in \text{Im } E$.

Given $\beta \in V$, $TE\beta \in \text{Im } E$, so $TE\beta = E\alpha$ for some $\alpha \in V$. Since E is idempotent,

$$ETE\beta = E^2\alpha = E\alpha = TE\beta, \text{ so shown.}$$

Conversely, given $\beta \in \text{Im } E$, $\beta = E\alpha$ for some $\alpha \in V$, and so

$$T\beta = TE\alpha = ETE\alpha \in \text{Im } E.$$

To show the second assertion, assume $\ker E$ and $\text{Im } E$ are invariant under T .

Given $\alpha \in V$, $\alpha = E\alpha + \alpha - E\alpha$. So,

#6.8, 4.

Let T be a linear operator on the fin. dim V with the characteristic χ and minimal μ

$$\chi(t) = (t - c_1)^{d_1} \cdots (t - c_k)^{d_k}$$

$$\mu(t) = (t - c_1)^{n_1} \cdots (t - c_k)^{n_k}$$

And $W_i = \ker(T - c_i I)^{n_i}$.

a). Prove that $W_i = \bigcup_{m \in \mathbb{N}} \ker(T - c_i I)^m$.

b). Prove that $\dim W_i = d_i$.

Proof. Clearly $W_i \subseteq \bigcup_{m \in \mathbb{N}} \ker(T - c_i I)^m$. Conversely, let $\alpha \in \bigcup_{m \in \mathbb{N}} \ker(T - c_i I)^m$ be given. That is, for some $m \in \mathbb{N}$, $(T - c_i I)^m(\alpha) = 0$.

Sec 17.2.

#16. Let $A \in M_{k \times k}(\mathbb{R})$ such that $A^2 + I = 0$. Then,

$$A \sim \begin{bmatrix} 0 & -I_k \\ I_k & 0 \end{bmatrix}.$$

Proof. Since $x^2 + 1$ is irreducible over $\mathbb{R}$, it is the minimal polynomial of A .

Moreover, the characteristic polynomial must be $(x^2 + 1)^k$, by the Cayley-Hamilton Thm.

Now, for $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in M_{2 \times 2}(\mathbb{R})$, A has a Rational form as

$$\begin{bmatrix} B & & 0 \\ & \ddots & \\ 0 & & B \end{bmatrix}.$$

Meanwhile, $\begin{bmatrix} 0 & -I_k \\ I_k & 0 \end{bmatrix} = R$ satisfies $R^2 + I = 0$, so it has same Rational form to A .

$A \sim R$ if the rational forms are same, therefore there are similar.

$$Re_1 = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \\ \vdots \end{pmatrix} = e_{k+1}, \quad Re_2 = e_{k+2}, \dots, Re_k = e_{2k}$$

$$Re_{k+1} = -e_1, \dots, Re_{2k} = -e_k.$$

FSNU. 2020 spring #2.

$$A = \begin{pmatrix} 0 & I_n \\ 2I_n & 0 \end{pmatrix} \in M_{2n}(\mathbb{R}).$$

a). A is diagonalizable.

b). Find characteristic and minimal polynomial of A .

Proof. $A^2 = \begin{pmatrix} 2I_n & 0 \\ 0 & 2I_n \end{pmatrix} = 2I_{2n}$, so $x^2 - 2$ is the minimal polynomial of A .

And, for $1 \leq i \leq n$, $Ae_i = 2 \cdot e_{i+n}$

for $n < i \leq 2n$, $Ae_i = e_{i-n}$

So, for $1 \leq i \leq n$,

$$A(ae_i + be_{i+n}) = 2a \cdot e_{i+n} + be_i$$

$$A(e_i + \sqrt{2} \cdot e_{i+n}) = 2 \cdot e_{i+n} + \sqrt{2} \cdot e_i = \sqrt{2}(e_i + \sqrt{2} e_{i+n})$$

$$\text{and } A(e_i - \sqrt{2} e_{i+n}) = 2e_{i+n} - \sqrt{2} e_i = -\sqrt{2}(e_i - \sqrt{2} e_{i+n})$$

$$\text{so, for } P = \begin{pmatrix} e_1 + \sqrt{2}e_{1+n} & \dots & e_n + \sqrt{2}e_{2n} & e_1 - \sqrt{2}e_{1+n} & \dots & e_n - \sqrt{2}e_{2n} \end{pmatrix},$$

$$P^{-1}AP = \begin{pmatrix} \sqrt{2}I_n & 0 \\ 0 & -\sqrt{2}I_n \end{pmatrix}$$

so, the characteristic polynomial is $(\sqrt{2} - t)^n \cdot (\sqrt{2} + t)^n$
 $= (2 - t^2)^n$

Sec. 8.1.

8.1.1. Let V be an Inn. space.

1). For all $\alpha \in V$, $\langle 0, \alpha \rangle = \langle \alpha, 0 \rangle = 0$.

2). If $\langle \alpha, \beta \rangle = 0$ for all $\alpha \in V$, then $\beta = 0$.

Proof. Given $\alpha \in V$, $\langle \alpha, 0 \rangle = \langle \alpha, 0 \cdot 0 \rangle = 0 \cdot \langle \alpha, 0 \rangle = 0$.

And, $\langle \beta, \beta \rangle = 0$ implies $\beta = 0$.

positive

8.1.2 Inner products are closed under the addition and scalar multiplication.

8.1.3. Describe explicitly all inner products on $\mathbb{R}^1$ and $\mathbb{C}^1$.

Proof. Given $\alpha, \beta \in \mathbb{R}$, $\langle \alpha, \beta \rangle = \alpha\beta \cdot \langle 1, 1 \rangle$. So, it is determined by the value of $\langle 1, 1 \rangle$.

So, the inner product is determined by the value of $\langle 1, 1 \rangle > 0$.

Given $a+ib, c+id \in \mathbb{C}$,

$$\begin{aligned}\langle a+ib, c+id \rangle &= \langle a, c+id \rangle + \langle ib, c+id \rangle \\ &= \langle a, c \rangle + \langle a, id \rangle + \langle ib, c \rangle + \langle ib, id \rangle\end{aligned}$$

8.1.5. Let $\langle, \rangle$ be the standard inner product on $\mathbb{R}^2$.

a). Let $\alpha = (1, 2)$ and $\beta = (-1, 1)$. Find $\gamma \in \mathbb{R}^2$ s.t.

$$\langle \alpha, \gamma \rangle = -1 \text{ and } \langle \beta, \gamma \rangle = 3.$$

Let $\gamma = a_1 e_1 + a_2 e_2$. Then, $\langle \alpha, \gamma \rangle = a_1 \cdot \langle \alpha, e_1 \rangle + a_2 \cdot \langle \alpha, e_2 \rangle = a_1 + 2a_2 = -1$

$$\langle \beta, \gamma \rangle = a_1 \cdot \langle \beta, e_1 \rangle + a_2 \cdot \langle \beta, e_2 \rangle = -a_1 + a_2 = 3$$

$$a_2 = \frac{2}{3}, a_1 = -\frac{17}{3}.$$

b). $d = \langle \alpha, e_1 \rangle e_1 + \langle \alpha, e_2 \rangle e_2$

Proof. $d = c_1 e_1 + c_2 e_2$ for some $c_1, c_2 \in \mathbb{R}$. Then,

$$\langle d, e_1 \rangle = c_1 \langle e_1, e_1 \rangle + 0 = c_1.$$

#8.2.1.

Let $W = \{(1, 0, -1, 1), (2, 3, -1, 2)\}^\perp$. Find a basis for W .

Proof. Let $W_1 = \text{span}_{\mathbb{R}}\{(1, 0, -1, 1), (2, 3, -1, 2)\}$.

$$\text{Then, } (a, b, c, d) \in W_1^\perp \Leftrightarrow \begin{cases} a + 0 - c + d = 0 \\ 2a + 3b - c + 2d = 0 \end{cases} \Rightarrow \begin{cases} 3b + c = a \\ 2a + 3b - c + 2d = 0 \end{cases}$$

$$\text{So, } W^\perp = \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -3 \\ 1 \\ -3 \\ 0 \end{pmatrix} \right\}$$

$$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} a \\ b \\ -3b \\ d \end{pmatrix} = \begin{pmatrix} c - d \\ b \\ -3b \\ d \end{pmatrix} = \begin{pmatrix} -3b - d \\ b \\ -3b \\ d \end{pmatrix} = b \begin{pmatrix} -3 \\ 1 \\ -3 \\ 0 \end{pmatrix} + d \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Recall: Row-Reduced Echelon Form.

$$\begin{pmatrix} 1 & 0 & -1 & 1 \\ 2 & 3 & -1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 3 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & \frac{1}{3} & 0 \end{pmatrix}$$

#8.2.2. Apply Gram-Schmidt to $\beta_1 = (1, 0, 1)$, $\beta_2 = (1, 0, -1)$, $\beta_3 = (0, 3, 4)$.

Take $\alpha_1 = \beta_1$,

$$\alpha_2 = \beta_2 - \frac{\langle \beta_2, \alpha_1 \rangle}{\|\alpha_1\|^2} \cdot \alpha_1 = (1, 0, -1) - 0,$$

$$\alpha_3 = \beta_3 - \frac{\langle \beta_3, \alpha_1 \rangle}{\|\alpha_1\|^2} \cdot \alpha_1 - \frac{\langle \beta_3, \alpha_2 \rangle}{\|\alpha_2\|^2} \cdot \alpha_2 = (0, 3, 4) - \frac{4}{2} \cdot (1, 0, 1) - \frac{-4}{2} \cdot (1, 0, -1)$$

$$= (0, 3, 4) - (2, 0, 2) + (2, 0, -2) = (0, 3, 0)$$

Now, $\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} \right\}$ is the desired.

#8.2.3. In $\mathbb{C}^3$, find orthonormal basis for $\text{span}_{\mathbb{C}}\left\{\begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 1+i \end{pmatrix}\right\}$.

Proof $\begin{pmatrix} 2 \\ 1 \\ 1+i \end{pmatrix} = \frac{\langle \begin{pmatrix} 2 \\ 1 \\ 1+i \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix} \rangle}{2} \cdot \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix} + \frac{1}{2} \cdot \begin{pmatrix} 1+i \\ 2 \\ 1-i \end{pmatrix}$

$$= \begin{pmatrix} 2 \\ 1 \\ 1+i \end{pmatrix} - \frac{2-i+1}{2} \cdot \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1+i \end{pmatrix} - \begin{pmatrix} \frac{3-i}{2} \\ 0 \\ \frac{1+i}{2} \end{pmatrix} = \begin{pmatrix} \frac{1+i}{2} \\ 1 \\ \frac{1-i}{2} \end{pmatrix}, \text{ so, } \left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1+i}{2} \\ 1 \\ \frac{1-i}{2} \end{pmatrix} \right\}.$$

#8.2.5. Let V be an inn. prod. space, and let $\alpha, \beta \in V$. Then,

$$\alpha = \beta \Leftrightarrow \langle \alpha, \gamma \rangle = \langle \beta, \gamma \rangle \text{ for all } \gamma \in V.$$

Proof. $\Rightarrow$ is clear, and $\langle \alpha - \beta, \gamma \rangle = 0$ for all $\gamma \in V$, then $\langle \alpha - \beta, \alpha - \beta \rangle = 0$, so $\alpha = \beta$.

#8.2.6. $E(x_1, x_2) = \frac{3x_1 + 4x_2}{25} \cdot \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ or $x_1 \cdot \begin{pmatrix} \frac{9}{25} \\ \frac{12}{25} \end{pmatrix} + x_2 \cdot \begin{pmatrix} \frac{12}{25} \\ \frac{16}{25} \end{pmatrix}$

$$E = \begin{pmatrix} \frac{9}{25} & \frac{12}{25} \\ \frac{12}{25} & \frac{16}{25} \end{pmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

And, $W^\perp = (\text{Im } E)^\perp = \text{Ker } E$. Clearly, $\text{Ker } E = \text{span}\left\{\begin{pmatrix} 4 \\ -3 \end{pmatrix}\right\}$ Now, for $A = \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$,

$$E = A \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} A^{-1}. \quad (A^{-1} = -\frac{1}{25} \cdot \begin{pmatrix} -3 & -4 \\ -4 & 3 \end{pmatrix})$$

#8.2.7. ?

#8.2.8. Find an inner product on $\mathbb{R}^2$ such that $\langle e_1, e_2 \rangle = 2$.

Proof In General, given $x, y \in F^n$ where $F = \mathbb{R}$ or $\mathbb{C}$, $\langle x, y \rangle$ is described by;
let $B = \{\alpha_1, \dots, \alpha_n\}$ be an ordered basis for F^n . Construct $A \in M_{n \times n}(F)$ as:

$$A_{ij} = \langle \alpha_j, \alpha_i \rangle. \quad (\text{so, } A \text{ is Hermitian, } \langle \alpha_j, \alpha_i \rangle = \overline{\langle \alpha_i, \alpha_j \rangle})$$

Then, $x = \sum x_i \alpha_i$ and $y = \sum y_i \alpha_i$,

$$\begin{aligned} \langle x, y \rangle &= \langle \sum x_i \alpha_i, y \rangle = \sum x_i \langle \alpha_i, y \rangle = \sum x_i \langle \alpha_i, \sum y_j \alpha_j \rangle = \sum x_i \cdot \sum \bar{y}_j \cdot \langle \alpha_i, \alpha_j \rangle \\ &= \sum_{i,j} \bar{y}_j \cdot A_{ji} \cdot x_i = y^* A x. \end{aligned}$$

Reversibly, if A is an invertible Hermitian st $x^* A x > 0$ for all $x \neq 0$, then $\langle x, y \rangle = y^* A x$ is an inner product. (Positive).

Let $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$. $e_2^* A e_1 = \left(e_2 \cdot \begin{pmatrix} a \\ b \end{pmatrix} \right) = b$, so, take $b=2$.

And, $(x_1 \ x_2) A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (x_1 \ x_2) \begin{pmatrix} ax_1 + 2bx_2 \\ 2bx_1 + dx_2 \end{pmatrix} = ax_1^2 + 2bx_1x_2 + dx_2^2 > 0$,

So take $a=5$, $d=1$.

#8.2.9. Let $V = P_3(\mathbb{R})$ with equipped an 'inner product

$$\langle f, g \rangle = \int_0^1 f(t) \cdot g(t) dt$$

a). Let $W = \{f \in V\}$. Find $W^\perp$.

b). Apply the Gram-Schmidt process to $\{1, x, x^2, x^3\}$.

Proof. Define a map $f \mapsto \int_0^1 f(t) dt \quad (= \frac{\langle f, 1 \rangle}{\|1\|^2} \cdot 1)$

Then, it is an orthogonal projection. So, the kernel of this map is $W^\perp$,

$$\{f \in P_3(\mathbb{R}) \mid \int_0^1 f dt = 0\}.$$

Let $f(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0$. Then,

$$\int_0^1 f dt = \frac{1}{4} a_3 t^4 + \frac{1}{3} a_2 t^3 + \frac{1}{2} a_1 t^2 + a_0 t \Big|_0^1 = \frac{1}{4} a_3 + \frac{1}{3} a_2 + \frac{1}{2} a_1 + a_0$$

Therefore, $W^\perp = \{f \in P_3(\mathbb{R}) \mid \int_0^1 f dt = 0\} = \text{span}_{\mathbb{R}} \left\{ x^3 - \frac{1}{4}, x^2 - \frac{1}{3}, x - \frac{1}{2} \right\}.$

$$x - \frac{\langle x, 1 \rangle}{\|1\|^2} \cdot 1 = x - \int_0^1 t dt = x - \frac{1}{2}$$

$$x^2 - \frac{\langle x^2, 1 \rangle}{\|1\|^2} \cdot 1 - \frac{\langle x^2, x - \frac{1}{2} \rangle}{\|x - \frac{1}{2}\|^2} \cdot (x - \frac{1}{2}) =$$

#8.2.11. Let V be a fin. dim inn. prod space and let $\{a_1, \dots, a_n\}$ be an orthonormal basis. For any $\alpha, \beta \in V$, $\langle \alpha, \beta \rangle = \sum_{k=1}^n \langle \alpha, a_k \rangle \cdot \overline{\langle \beta, a_k \rangle}$

Proof. Ask...

$$\alpha = \langle \alpha, a_1 \rangle \cdot a_1 + \dots + \langle \alpha, a_k \rangle \cdot a_k$$

$$\beta = \langle \beta, a_1 \rangle \cdot a_1 + \dots + \langle \beta, a_k \rangle \cdot a_k$$

$$\begin{aligned} \text{So, } \langle \alpha, \beta \rangle &= \sum_{i=1}^n \langle \alpha, a_i \rangle \cdot \overline{\langle \beta, a_i \rangle} = \sum_{i=1}^n \langle \alpha, a_i \rangle \cdot \overline{0 + \dots + \langle \beta, a_i \rangle + \dots + 0} \\ &= \sum_{i=1}^n \langle \alpha, a_i \rangle \cdot \overline{\langle \beta, a_i \rangle} \end{aligned}$$

#8.2.12. Let W be a finite-dimensional subspace of V , and let E be the orthogonal projection of V on W .

Prove that for all $\alpha, \beta \in V$, $\langle E(\alpha), \beta \rangle = \langle \alpha, E(\beta) \rangle$.

Proof. Since E is orthogonal projection, $\text{Im } E = W$, and $\text{Ker } E = W^\perp$ with $\text{Im } E \oplus \text{Ker } E$. Given $\alpha, \beta \in V$, there are $\alpha_1, \beta_1 \in \text{Im } E$ and $\alpha_2, \beta_2 \in \text{Ker } E$ such that

$$\alpha = \alpha_1 + \alpha_2 \quad \text{and} \quad \beta = \beta_1 + \beta_2$$

$$\text{So, } \langle E(\alpha), \beta \rangle = \langle \alpha_1, \beta_1 + \beta_2 \rangle = \langle \alpha_1, \beta_1 \rangle$$

$$\text{and } \langle \alpha, E(\beta) \rangle = \langle \alpha_1 + \alpha_2, \beta_1 \rangle = \langle \alpha_1, \beta_1 \rangle$$

Thus, E is a Hermitian.

8.2.13.

Let S be a subset of an inn. prod space V . Then,

1). $S^\perp = \langle S \rangle^\perp$ and $S^\perp$ is subspace

2). $W \subseteq (W^\perp)^\perp$.

3). If W is fin. dim., then $W = (W^\perp)^\perp$
 $\xrightarrow{S \subseteq \langle S \rangle}$
 Proof. Given $\alpha \in V$,

$\alpha \in S^\perp \Leftrightarrow$ for any $\mu \in S$, $\langle \alpha, \mu \rangle = 0 \Leftrightarrow$ for any $\mu \in \langle S \rangle$, $\langle \alpha, \mu \rangle = 0 \Leftrightarrow \alpha \in \langle S \rangle^\perp$.
 $\xrightarrow{\text{linearity of inn. prod.}}$ 1)

Given $\alpha \in W$, $\langle \alpha, \mu \rangle = 0$ for all $\mu \in W^\perp$, so $\alpha \in (W^\perp)^\perp$.

But the converse is not true: Let $V = \mathbb{R}[x]$ and define an inn. prod. on V as

$$\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}; \left(\sum_{i=1}^n a_i x^i, \sum_{i=1}^m b_i x^i \right) \mapsto \sum_{i=1}^{n+m} a_i \overline{b_i}.$$

And $W = \langle \{1+x^n \mid n \in \mathbb{N}\} \rangle$. Then, $W \neq V$ and $W^\perp = \{0\}$, so $(W^\perp)^\perp = V$. 12)

Let W be a fin. dim. Then, there exists an orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$ for W .

Given $\alpha \in (W^\perp)^\perp$, $\langle \alpha, \mu \rangle = 0$ for all $\mu \in W^\perp$.

Suppose that $\alpha \notin W$. Then, there is an orthogonal projection $P(\alpha)$ on W .

Namely, $\alpha = P(\alpha) + (\alpha - P(\alpha))$, $\alpha - P(\alpha) \in W^\perp$, non-zero by $\alpha \notin W$.

But, $\langle \alpha, \alpha - P(\alpha) \rangle = \langle P(\alpha) + (\alpha - P(\alpha)), \alpha - P(\alpha) \rangle = 0 + \|\alpha - P(\alpha)\|^2 > 0$, so

for some $\mu \in W^\perp$ ($\mu = \alpha - P(\alpha)$), $\langle \alpha, \mu \rangle \neq 0$. It is Contradiction.

#8.2.14. Let V be a fin. dim inn. prod space, and let $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis.

Let $A = [T]_{\mathcal{B}}$. Then, $A_{i,j} = \langle T(\alpha_j), \alpha_i \rangle$

Proof. Given $1 \leq i \leq n$, $T(\alpha_j) = \langle T(\alpha_j), \alpha_1 \rangle \cdot \alpha_1 + \dots + \langle T(\alpha_j), \alpha_n \rangle \cdot \alpha_n$

Therefore,

$$A = \begin{pmatrix} [T(\alpha_1)]_{\mathcal{B}} & \dots & [T(\alpha_n)]_{\mathcal{B}} \end{pmatrix} = \begin{pmatrix} \langle T(\alpha_1), \alpha_1 \rangle & & \\ \langle T(\alpha_1), \alpha_2 \rangle & \ddots & \\ \vdots & & \\ \langle T(\alpha_1), \alpha_n \rangle & & \end{pmatrix}.$$

#8.2.15.

Suppose $V = W_1 \oplus W_2$ and $f_i: W_i \times W_i \rightarrow F$ ($i=1,2$) are inner product on W_1 and W_2 , respectively, where $F = \mathbb{R}$ or $\mathbb{C}$. Show that there exists a unique inner product f on V s.t

a) $W_2 = W_1^\perp$

b) $f(\alpha, \beta) = f_k(\alpha, \beta)$ when $\alpha, \beta \in W_k$, $k=1,2$

#8.2.16.

Let V be an inner product space and W is fin. dim subspace.

If a projection $E: V \rightarrow V$ with $\text{Im } E = W$ satisfies

$$\|E(\alpha)\| \leq \|\alpha\| \quad \text{for all } \alpha \in V,$$

then E is orthogonal projection on W .

#8.2.17. Let $V = \{f: [-1, 1] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$ with equipped an inner product

$$\langle f, g \rangle = \int_{-1}^1 f(t) \cdot g(t) dt$$

Let W be the subspace of odd functions. Find $W^\perp$.

Proof. $f \in W^\perp \Leftrightarrow$ for any odd function g , $\int_{-1}^1 f(t) g(t) dt = 0$

Since for any even function g and any odd f , fg is odd, so $\int_{-1}^1 fg = 0$.

That is, $W^\perp$ contains all even function. Moreover, for any $f \in V$,

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd}}$$

So, for subset W' of V consisting by all even functions,

$$V = W + W', \text{ and } W' \subseteq W^\perp.$$

Now, clearly $W \cap W' = \{0\}$, $V = W \oplus W'$.

If $W' \neq W^\perp$, then, for some $h \in W^\perp$, $h \notin W'$. Again, h can be decomposed as

$$h(x) = \underbrace{\frac{h(x) + h(-x)}{2}}_{\in W'} + \underbrace{\frac{h(x) - h(-x)}{2}}_{\in W}$$

Since $W' \subseteq W^\perp$, $h(x) - \frac{h(x) + h(-x)}{2} \in W^\perp$, and $\in W$.

But, this implies that it is contained in $W \cap W^\perp$, so it is zero:

$$\text{If } x \in W \cap W^\perp, \text{ then } \langle x, x \rangle = 0, \text{ or } x = 0.$$

Thus, $h(x) - \frac{h(x) + h(-x)}{2} = \frac{h(x) - h(-x)}{2} = 0$, so $h(x) = h(-x)$, even. Contradicts to $h \notin W'$.

Now, we can construct a projection of V onto W as:

$$P: V \rightarrow V: f(x) \mapsto \frac{f(x) - f(-x)}{2}$$

8.3 Adjoint.

#8.3.1. In $\mathbb{C}^2$, equipped the standard inn. prod,
let T be the linear operation defined by $Te_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$, $Te_2 = \begin{pmatrix} i \\ -1 \end{pmatrix}$.
Find a formula $T^*(x_1, x_2)$.

Proof Since $\{e_1, e_2\}$ is orthonormal basis for $\mathbb{C}^2$,

$$[T^*]_{\mathcal{B}} = [T]_{\mathcal{B}}^* = \begin{pmatrix} 1 & i \\ -2 & -1 \end{pmatrix}^* = \begin{pmatrix} 1 & -2 \\ -i & -1 \end{pmatrix}.$$

$$\text{So, } T^*(x_1, x_2) = \begin{pmatrix} 1 & -2 \\ -i & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 - 2x_2 \\ -ix_1 - x_2 \end{pmatrix}.$$

#8.3.2. $[T]_{\mathcal{B}} = \begin{pmatrix} 1+i & i \\ 2 & i \end{pmatrix}$, $[T^*]_{\mathcal{B}} = \begin{pmatrix} 1-i & 2 \\ -i & -i \end{pmatrix}$

Commutate?

$$T \cdot T^* = \begin{pmatrix} 3 & \end{pmatrix} \neq T^* T = \begin{pmatrix} 6 & \end{pmatrix}$$

No!

#8.3.3. $T = \begin{bmatrix} -1 & -2 & 1 \\ -2 & 1 & 2 \\ 1 & i & -1 \end{bmatrix}$, $T^* = \begin{bmatrix} -1 & i & 1 \\ i & 1 & -2 \\ 1 & -i & -1 \end{bmatrix}$

#8.3.4

Let V be a fin. dim inn. prod space, and $T: V \rightarrow V$ be a linear operator.
Then, $\text{Im } T^* = (\text{Ker } T)^\perp$.

Proof. Let $T^*(\alpha) \in \text{Im } T^*$ be given. Then, for any $\beta \in \text{Ker } T$,

$$\langle T^*(\alpha), \beta \rangle = \langle \alpha, T(\beta) \rangle = \langle \alpha, 0 \rangle = 0, \text{ so } \text{Im } T^* \subseteq (\text{Ker } T)^\perp.$$

Conversely, since V is finite dimensional,

$$((\text{Ker } T)^\perp)^\perp = \text{Ker } T$$

$$\text{and } \alpha \in \text{Ker } T \Leftrightarrow \text{for all } \beta \in V, 0 = \langle \alpha, \beta \rangle = \langle T(\alpha), \beta \rangle = \langle \alpha, T^*(\beta) \rangle \\ \Leftrightarrow \alpha \in (\text{Im } T^*)^\perp.$$

$$\text{So, } \text{Ker } T = (\text{Im } T^*)^\perp, \text{ so } (\text{Ker } T)^\perp = ((\text{Im } T^*)^\perp)^\perp = \text{Im } T^*.$$

#8.3.5.

#8.3.6. Let V be an inner product space, and β, γ be fixed vectors in V .
 Show that $T: V \rightarrow V; \alpha \mapsto \langle \alpha, \beta \rangle \cdot \gamma$

is a linear operator, and it has an adjoint. Describe explicitly the adjoint.

Proof. 1). T is linear. Given $\alpha_1, \alpha_2 \in V$ and $c \in F$,

$$T(c\alpha_1 + \alpha_2) = \langle c\alpha_1 + \alpha_2, \beta \rangle \cdot \gamma = (c \cdot \langle \alpha_1, \beta \rangle + \langle \alpha_2, \beta \rangle) \cdot \gamma = c \cdot T(\alpha_1) + T(\alpha_2).$$

2). T has an adjoint.

Given $\alpha_1, \alpha_2 \in V$,

$$\langle T(\alpha_1), \alpha_2 \rangle = \langle \langle \alpha_1, \beta \rangle \cdot \gamma, \alpha_2 \rangle = \langle \alpha_1, \beta \rangle \cdot \langle \gamma, \alpha_2 \rangle = \langle \gamma, \alpha_2 \rangle \cdot \langle \alpha_1, \beta \rangle = \langle \alpha_1, \langle \alpha_2, \gamma \rangle \cdot \beta \rangle$$

So, $T^*(\alpha) = \langle \alpha, \gamma \rangle \cdot \beta$ is the adjoint of T .

Particularly, let $\mathbb{C}^n$ with standard 'inn. Prod. put

$$\beta = [\gamma_1, \dots, \gamma_n] \quad \text{and} \quad \gamma = [x_1, \dots, x_n]$$

$$T(e_i) = \langle e_i, \beta \rangle \cdot \gamma = \bar{\gamma}_i \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Then, for standard basis e_i for $\mathbb{C}^n$,

$$[T]_{\mathcal{B}} = \begin{pmatrix} [T(e_1)]_{\mathcal{B}} & \dots & [T(e_n)]_{\mathcal{B}} \end{pmatrix} = \begin{pmatrix} x_1 \bar{\gamma}_1 & & \\ x_2 \bar{\gamma}_1 & \ddots & \\ \vdots & & \\ x_n \bar{\gamma}_1 & \dots & x_n \bar{\gamma}_n \end{pmatrix} = (x_i \bar{\gamma}_j)_{i,j}$$

The rank of $[T]_{\mathcal{B}}$ is exactly 1, if neither β nor γ are non-zero.

Note: It is element of the Singular value decomposition.

8.3.8,

3, 4 Let $V = P_3(\mathbb{R})$ with inn. prod $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$
For a fixed $t \in \mathbb{R}$, find $g_t \in V$ s.t.

$$\langle f, g_t \rangle = f(t) \text{ for all } f \in V.$$

And, for differentiation D , find D^*g .

Proof. Since $V \rightarrow \mathbb{R}; f \mapsto f(t)$ is a linear functional,
the Riesz-representation Thm gives that for some $g_t \in V$, $\langle \cdot, g_t \rangle = f \mapsto f(t)$.
More specifically, take $\mathcal{B} = \{\alpha_1, \dots, \alpha_4\}$ as orthonormal basis for V .

Then, for $g_t = \alpha_1(t) \cdot \alpha_1 + \alpha_2(t) \cdot \alpha_2 + \alpha_3(t) \cdot \alpha_3 + \alpha_4(t) \cdot \alpha_4$,

$$\langle f, g_t \rangle = \sum \alpha_i(t) \cdot \langle f, \alpha_i \rangle = f(t).$$

Construct an orthonormal basis as: for $\beta = \{1, x, x^2, x^3\}$,

$$\begin{aligned} \alpha_1 &= 1. \\ d_2 &= x - \frac{\int_0^1 x \cdot 1 dx}{\|1\|^2} \cdot 1 = x - \frac{1}{2}. \\ d_3 &= x^2 - \frac{\int_0^1 x^2 \cdot 1 dx}{\|1\|^2} \cdot 1 - \frac{\int_0^1 x^2 \cdot (x - \frac{1}{2}) dx}{\|x - \frac{1}{2}\|^2} \cdot (x - \frac{1}{2}) = x^2 - 2x^2 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{1}{36} \\ &= x^2 - \frac{1}{3} - (x - \frac{1}{2}) = x^2 - x + \frac{1}{6}. \\ d_4 &= x^3 - \frac{\int_0^1 x^3 \cdot 1 dx}{\|1\|^2} \cdot 1 - \frac{\int_0^1 x^3 \cdot (x - \frac{1}{2}) dx}{\|x - \frac{1}{2}\|^2} \cdot (x - \frac{1}{2}) - \frac{\int_0^1 x^3 (x^2 - x + \frac{1}{6}) dx}{\|(x^2 - x + \frac{1}{6})\|^2} \cdot (x^2 - x + \frac{1}{6}) \\ &= x^3 - \frac{3}{2}x^2 - \frac{12}{5}x - \frac{1}{20}. \end{aligned}$$

And, processing Gram-Schmidt process we can obtain the

adjoint of

And, find the differentiation: $\langle Df, g \rangle = \langle f, D^*g \rangle$, so

$$\langle Df, g \rangle = \int_0^1 Df(x)g(x) dx = \int_0^1 D(f(x)g(x)) dx - \int_0^1 f(x)Dg(x) dx = \langle D(fg), 1 \rangle - \langle f, Dg \rangle$$

It equals to $\langle f, D^*g \rangle$, so

$$\langle f, D^*g + Dg \rangle = \langle f, (D^* + D)(g) \rangle = \langle D(fg), 1 \rangle = f(0)g(0) - f(1)g(0)$$

Let $g = 1$.

#8.3.10. Let $V = M_{n \times n}(\mathbb{C})$, with an inner product

$$\langle A, B \rangle = \text{trace}(AB^*).$$

Let P be a fixed invertible matrix in V , and $T_P(A) = P^{-1}AP$. Find T_P^* .

Proof. Given $A, B \in V$,

$$\begin{aligned} \langle T_P(A), B \rangle &= \langle P^{-1}AP, B \rangle = \text{tr}(P^{-1}APB^*) = \text{tr}(APB^*P^{-1}) = \langle A, (PB^*P^{-1})^* \rangle \\ &= \langle A, (P^*)^{-1}B^*P^* \rangle = \langle A, T_{P^*}(B) \rangle \end{aligned}$$

$$\text{so, } (T_P)^* = T_{P^*}.$$

#8.3.11 Let V be a fin. dim inn. prod space, and E be an idempotent linear map on V . Then, E is Hermitian iff E is Normal.

Proof. If Hermitian, then clearly Normal. (Not necessary idempotent).

If E is Normal, that is, $EE^* = E^*E$, then given $\alpha, \beta \in V$,

$$\langle EE^*(\alpha), \beta \rangle = \langle E^*(\alpha), E^*(\beta) \rangle \stackrel{\text{normal}}{=} \langle E^*E(\alpha), \beta \rangle = \langle E(\alpha), E(\beta) \rangle$$

So, $\|E^*(\alpha)\| = 0 \Leftrightarrow \|E(\alpha)\| = 0$, or $\ker E^* = \ker E$. $\perp \ker E = \ker E^*$ only needed Normal

Since E is an idempotent, for any $x \in V$, $x - E(x) \in \ker E = \ker E^*$,

so $E^*(x - E(x)) = E^*(x) - (E^* \circ E)(x) = 0$, thus $E^*(x) = (E^* \circ E)(x) = (E \circ E^*)(x)$.

That is, $E^* = E^* \circ E$. Since $(E^* \circ E)^* = E^* \circ E^{**} = E^* \circ E$, $E^* = E$. $\square$

#8, 3, 12. Let V be a fin. dim Complex inn. prod space, and let T be a linear operator on V . Then, T is Hermitian if and only if for all $\alpha \in V$, $\langle T(\alpha), \alpha \rangle \in \mathbb{R}$.

Proof. If T is hermitian, given $\alpha \in V$, $\langle T(\alpha), \alpha \rangle = \langle \alpha, T^*(\alpha) \rangle = \langle \alpha, T(\alpha) \rangle = \overline{\langle T(\alpha), \alpha \rangle}$, so it is real-valued. To show the converse, we will show that: $\langle T(\alpha), \alpha \rangle = 0$ for all $\alpha \in V$, then $T = T^*$. Given $x, y \in V$,

$$0 = \langle T(x+y), x+y \rangle = \langle T(x), x \rangle + \langle T(y), y \rangle + \langle T(x), y \rangle + \langle T(y), x \rangle = \langle T(x), y \rangle + \langle T(y), x \rangle$$

and

$$0 = \langle T(x+iy), x+iy \rangle = \langle T(x), iy \rangle + \langle T(iy), x \rangle = -i \langle T(x), y \rangle + i \langle T(y), x \rangle$$

$$\text{Thus, } 0 = -\langle T(x), y \rangle + \langle T(y), x \rangle$$

Combining the above two results, then $\langle T(x), y \rangle = 0$, for all $x, y \in V$.

Take $y = T(x)$, then $\|T(x)\|^2 = 0$ for all x , so $T = 0$.

Now, observe that $T = T^* \Leftrightarrow T - T^* = 0$.

If for all $\alpha \in V$, $\langle T(\alpha), \alpha \rangle \in \mathbb{R}$,

$$\langle T(\alpha), \alpha \rangle = \overline{\langle \alpha, T(\alpha) \rangle} = \langle \alpha, T(\alpha) \rangle = \langle T^*(\alpha), \alpha \rangle$$

So, $\langle (T - T^*)(\alpha), \alpha \rangle = 0$, it implies $T - T^* = 0$, or T is Hermitian.

Section. 8.4 Unitary operator.

#1. Ask a teacher

#2. Let $V = M_n(\mathbb{C})$ with an inner product

$$\langle A, B \rangle = \text{tr}(AB^*).$$

For each $M \in V$, define $T_M(A) = MA$. Show that

T_M is unitary if and only if M is unitary.

Proof. T_M is unitary $\Leftrightarrow$ for all A and B in V , $\langle MA, MB \rangle = \langle A, B \rangle$.

$$\Leftrightarrow \text{tr}(MAB^*M^*) = \text{tr}(AB^*)$$

Since $\text{tr}(MAB^*M^*) = \text{tr}(M^*MA B)$, if M is unitary,
 $= \text{tr}(AB^*)$, so T_M is unitary.

And,

#3. Let $V = \mathbb{C}$ be the vector space over $\mathbb{R}$.

a). $\langle \alpha, \beta \rangle = \operatorname{Re}(\alpha \bar{\beta})$ is an inner product on V .

Proof. Given $\alpha, \beta, \gamma \in \mathbb{C}$ and $c \in \mathbb{R}$,

$$\langle c\alpha + \beta, \gamma \rangle = \operatorname{Re}([c\alpha + \beta] \cdot \bar{\gamma}) = \operatorname{Re}(c \cdot \alpha \cdot \bar{\gamma} + \beta \bar{\gamma}) = c \cdot \operatorname{Re}(\alpha \cdot \bar{\gamma}) + \operatorname{Re}(\beta \bar{\gamma})$$

$$= c \cdot \langle \alpha, \gamma \rangle + \langle \beta, \gamma \rangle. \quad \text{Linearity}$$

$$\langle \alpha, \beta \rangle = \operatorname{Re}(\alpha \bar{\beta}) = \operatorname{Re}(\overline{\beta \alpha}) = \operatorname{Re}(\overline{\beta} \alpha) = \operatorname{Re}(\alpha \bar{\beta}) = \langle \beta, \alpha \rangle \quad \text{Conjugate}$$

$$\langle \alpha, \alpha \rangle = \operatorname{Re}(\alpha \bar{\alpha}) = \operatorname{Re}(|\alpha|^2) \geq 0 \quad \text{iff } \alpha \neq 0 \quad \text{Positive}$$

b). Find inner product preserved isomorphism onto $\mathbb{R}^2$ with Standard inn. Prod.

Proof. Clearly, $\mathcal{B}_2 = \{e_1, e_2\}$ is orthonormal basis for $\mathbb{R}^2$.

And, $\mathcal{B}_1 = \{1, i\}$ is an orthonormal basis for $\mathbb{C}$ over $\mathbb{R}$.

$$\langle 1, 1 \rangle = 1, \quad \langle i, i \rangle = \operatorname{Re}(i \cdot -i) = \operatorname{Re}(1) = 1, \quad \langle 1, i \rangle = \operatorname{Re}(-i) = 0$$

Now, define $T: \mathbb{C} \rightarrow \mathbb{R}^2$ s.t. $T(1) = e_1, \quad T(i) = e_2$.

$$\text{Then is, } [T]_{\mathcal{B}_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \text{ or } T(a+bi) = \begin{pmatrix} a \\ b \end{pmatrix}.$$

c). For each $\gamma \in V$, define $M_\gamma: V \rightarrow V$ s.t. $\gamma \alpha$.

Show that $(M_\gamma)^* = M_{\bar{\gamma}}$

Proof. Given $\alpha, \beta \in V$,

$$\langle M_\gamma(\alpha), \beta \rangle = \langle \gamma \alpha, \beta \rangle = \operatorname{Re}(\gamma \alpha \bar{\beta}) = \langle \alpha, \beta \bar{\gamma} \rangle = \langle \alpha, M_\gamma^*(\beta) \rangle$$

Since α and β were arbitrary, $M_\gamma^*(\beta) = \beta \bar{\gamma} = M_{\bar{\gamma}}(\beta)$.

td).

So, $M_\gamma = M_\gamma^* \Leftrightarrow \gamma \alpha = \bar{\gamma} \alpha$ for all $\alpha \in \mathbb{C}$, so $\gamma = \bar{\gamma}$, or γ be real.

#7. Let V be a Complex inner product space, and T be a Hermitian on V .
Show that

a) $\|\alpha + iT(\alpha)\| = \|\alpha - iT(\alpha)\|$ for all $\alpha \in V$

b) $\alpha + iT(\alpha) = \beta + iT(\beta)$ iff $\alpha = \beta$.

c) $I \pm iT$ is non-singular.

d) If V is fin. dim, then $U = (I - iT)(I + iT)^{-1}$ is unitary.

Proof. $\langle \alpha + iT(\alpha), \alpha + iT(\alpha) \rangle = \|\alpha\|^2 + \langle \alpha, iT(\alpha) \rangle + \langle iT(\alpha), \alpha \rangle + \|iT(\alpha)\|^2$

$$\langle \alpha - iT(\alpha), \alpha - iT(\alpha) \rangle = \|\alpha\|^2 - (\langle \alpha, iT(\alpha) \rangle + \langle iT(\alpha), \alpha \rangle) + \|iT(\alpha)\|^2.$$

And, since T is Hermitian, $\langle \alpha, iT(\alpha) \rangle = -i \cdot \langle T(\alpha), \alpha \rangle = -\langle iT(\alpha), \alpha \rangle$

and $\langle iT(\alpha), \alpha \rangle = i \langle \alpha, T(\alpha) \rangle = -\langle \alpha, iT(\alpha) \rangle$

so these are same.

b). $\alpha + iT(\alpha) = \beta + iT(\beta) \Rightarrow \alpha - \beta = i \cdot T(\beta - \alpha)$

$$\Rightarrow \langle T(\beta - \alpha), \beta - \alpha \rangle = \langle i(\beta - \alpha), \beta - \alpha \rangle = i \cdot \|\beta - \alpha\|^2.$$

Since if T Hermitian $\Rightarrow \langle T(\alpha), \alpha \rangle \in \mathbb{R}$, $\beta = \alpha$. The converse is trivial.

$$(\langle T(\alpha), \alpha \rangle = \langle \alpha, T(\alpha) \rangle = \overline{\langle T(\alpha), \alpha \rangle})$$

c). If $(I + iT)(\alpha) = 0$, then $iT(\alpha) = -\alpha \Rightarrow T(\alpha) = i\alpha$.

That is, $\langle T(\alpha), \alpha \rangle = i \cdot \|\alpha\|^2$, so α must be zero.

d). By c). in fin. dim, the $I \pm iT$ are invertible. Given $\alpha \in V$, let $\beta = (I + iT)^{-1}(\alpha)$.
Then, $\alpha = (I + iT)(\beta)$,

$$\|U(\alpha)\| = \|(I - iT)(\beta)\| = \|(I + iT)(\beta)\| = \|\alpha\|$$

So, it preserves the norm, so Unitary.



Normal and Hermitian.

Lemma. Let V be an inn. prod space and T is a linear operator.

(If exists) Every eigenvalue of T is real.

Moreover, eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof. Given eigenvalue C of T corresponding to α ,

$$C \cdot \langle \alpha, \alpha \rangle = \langle T(\alpha), \alpha \rangle = \langle \alpha, T^*(\alpha) \rangle = \langle \alpha, T(\alpha) \rangle = \langle \alpha, C \cdot \alpha \rangle = \bar{C} \cdot \langle \alpha, \alpha \rangle$$

Since α is non-zero, $C = \bar{C}$, or C is a real number.

And, take α_1, α_2 be eigenvectors corresponding to C_1, C_2 , with $C_1 \neq C_2$.

Then,

$$C_1 \cdot \langle \alpha_1, \alpha_2 \rangle = \langle C_1 \alpha_1, \alpha_2 \rangle = \langle T(\alpha_1), \alpha_2 \rangle = \langle \alpha_1, T^*(\alpha_2) \rangle = \langle \alpha_1, T(\alpha_2) \rangle = \langle \alpha_1, C_2 \alpha_2 \rangle = \bar{C}_2 \cdot \langle \alpha_1, \alpha_2 \rangle$$

Since C_2 must be real, $\bar{C}_2 = C_2$. Now, since $C_1 \neq C_2$,

$$(C_1 - C_2) \cdot \langle \alpha_1, \alpha_2 \rangle = 0 \text{ implies } \langle \alpha_1, \alpha_2 \rangle = 0.$$

Thm. In finite-dimensional inn. prod space,

Every Hermitian operator has an eigenvector.

Proof. If the given field is $\mathbb{C}$, then the characteristic polynomial must be a root in $\mathbb{C}$ by the Fundamental Theorem of Algebra. So it has an eigenvector.

Now, assume that the given field is the $\mathbb{R}$.

Let T be a Hermitian on V over $\mathbb{R}$, and β be an orthonormal basis for V .

Let $A = [T]_{\beta}$. Since $T = T^* = T^t$, $A = A^* = A^t$. Define an inner product on $\mathbb{C}^n$ as:

$$\langle X, Y \rangle \stackrel{\text{def}}{=} Y^* X$$

Then, a linear operator $L_A: \mathbb{C}^n \rightarrow \mathbb{C}^n: X \mapsto AX$ is Hermitian, because

$$\langle L_A(X), Y \rangle = \langle X, L_A^*(Y) \rangle \text{ and } \langle AX, Y \rangle = Y^* AX = \langle X, A^* Y \rangle = \langle X, AY \rangle.$$

By the FTA, L_A has an eigenvalue, and that must be real, by the above Lemma.

Now, $L_A(X) = AX = cX$ for some real number c and a non-zero vector X in $\mathbb{C}^n$.

or $\det(cI - A) = 0$, $A - cI$ is not invertible, and $A - cI$ has only real entries,

so $(A - cI)X = 0$ has a $\underbrace{\text{non-zero}}_{\text{non-zero}}$ solution X , so proved.

Sec. 8.5.

#1. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$\det(A - tI) = (t-1)^2 - 1 = t(t-2)$, so

let $P = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$, then $P^{-1} = \frac{1}{1} \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

$P^{-1}AP = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}}$

$= \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$

$\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$

$\det(tI - A) = (t^2 - \cos^2 \theta) - \sin^2 \theta = t^2 - 1 = (t-1)(t+1)$

$\begin{bmatrix} \cos \theta + 1 & \sin \theta \\ \sin \theta & -\cos \theta + 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{pmatrix} x \cos \theta + y \sin \theta + x \\ x \sin \theta - y \cos \theta + y \end{pmatrix} = 0.$

$\begin{cases} x(\cos \theta + 1) + y \sin \theta = 0 \\ y(1 - \cos \theta) + x \sin \theta = 0 \end{cases}$

$y = \frac{\sin \theta}{1 - \cos \theta}$, $x \cdot (\cos \theta + 1) = \sin \theta \cdot \frac{\sin \theta}{\cos \theta - 1} = \frac{1 - \cos^2 \theta}{\cos \theta - 1} = -(1 + \cos \theta)$

so $x = -1$, $\begin{pmatrix} \cos \theta - 1 \\ \sin \theta \end{pmatrix}$

$P = \begin{pmatrix} \cos \theta - 1 \\ \sin \theta \end{pmatrix}$

$\sin^2 \theta + \cos^2 \theta - 2\cos \theta + 1$
 $= 2 - 2\cos \theta$

$$\#3. A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

Since the characteristic polynomial is invariant by similar,

$$\det(tI - A) = \begin{vmatrix} t-1 & -2 & -3 \\ -2 & t-3 & -4 \\ -3 & -4 & t-5 \end{vmatrix}$$